

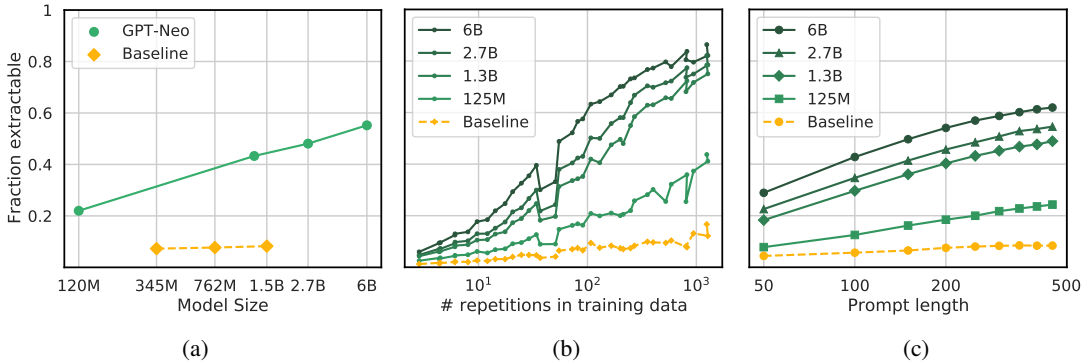


Figure 1: We prompt various sizes of GPT-Neo models (green) with data sampled from their training set—The Pile, and normalized by sequence lengths and duplication counts. As a baseline (yellow), we also prompt the GPT-2 family of models with the same Pile-derived prompts, even though these models were trained on WebText, a different training dataset. **(a)** Larger models memorize a larger fraction of their training dataset, following a log-linear relationship. This is not just a result of better generalization, as shown by the lack of growth for the GPT-2 baseline models. **(b)** Examples that are repeated more often in the training set are more likely to be extractable, again following a log-linear trend (baseline is GPT-2 XL). **(c)** As the number of tokens of context available increases, so does our ability to extract memorized text (baseline is GPT-2 XL).

the first $\ell - 50$ tokens and report the sequence as “extractable” if the model exactly emits the next 50 token suffix of this sequence. Fifty tokens corresponds to an average of 127 characters or 25 words in the GPT-Neo training set, well over the length of a typical English sentence. Finally, we compute the average probability that a sequence is extractable by averaging over all lengths ℓ .

4 EXPERIMENTS

We primarily study the GPT-Neo model family (Black et al., 2021; Wang and Komatsuzaki, 2021) trained on the Pile dataset (Gao et al., 2020). The GPT-Neo models are causal language models trained with the objective of predicting the next token in a sequence given the previous ones. They come in four sizes: 125 million, 1.3 billion, 2.7 billion and 6 billion parameters.¹ The Pile is a dataset of 825GB of text collected from various sources (e.g., books, Web scrapes, open source code). Prior to the recent release of OPT (Zhang et al., 2022), the GPT-Neo models were the largest language models available for public download, and The Pile is the largest public text dataset available.

4.1 BIGGER MODELS MEMORIZE MORE

We begin by considering the impact of model size on memorization, expanding on prior studies which qualitatively established a relationship between the size of GPT-2 models and their ability to memorize <30 URLs (Carlini et al., 2020). In contrast, we study *a million* model generations in order to describe how model scale relates to memorization.

Results. We first study our biased random data sample normalized by duplication count and sequence lengths. The results of this experiment are given in Figure 1a. The y-axis reports the fraction of generations which exactly reproduce the true suffix for their prompt, averaged over all prompt and sequence lengths in our evaluation set. Because our biased sampling over-represents duplicated strings, the *absolute* degree of memorization in Figure 1a is not particularly important here—rather, we are interested in how memorization varies with scale.² We find that larger models memorize significantly more than smaller models do, with a *near-perfect log-linear fit* (R^2 of 99.8%): a ten fold increase in model size corresponds to an increase in memorization of 19 percentage points.

¹As of February 2022, there is also a 20 billion parameter variant. Unfortunately this model uses a different training setup and tokenizer making it difficult to apply here.

²We repeat this experiment for a uniformly random subset of the data in Figure 2a.

To confirm that larger models are indeed *memorizing* more data, and not simply *generalizing* better, we repeat the analysis with the GPT-2 model family as a baseline. The GPT-2 models are similarly sized, and also trained on Internet-scraped data. If our “larger models memorize more” result was due to the predictive strength of larger models, and not the memorization of specific training data, we would expect a similar relationship between comparably sized GPT-2 models trained on similar data. Put differently, this baseline allows to establish what fraction of the training data is sufficiently “easy” that any language model can correctly predict the 50-token suffix, even if the example has not been seen during training. For example, a language model trained on multiple examples of number sequences can likely correctly complete some other unseen number sequences. We find that GPT-2 correctly completes approximately 6% of the examples in our evaluation set, compared to 40% for the similarly sized 1.3B parameter GPT-Neo model. A qualitative analysis (see examples in Appendix Figure 15) suggests that examples “memorized” by GPT-2 are largely uninteresting sequences (e.g., number sequences, repetitions of the same few tokens, or common phrases). Therefore, we conclude that larger models have a higher fraction of extractable training data because they have actually memorized the data; it is not simply that the larger models are more accurate.

4.2 REPEATED STRINGS ARE MEMORIZED MORE

Prior work provides preliminary evidence that memorization in language models increases with the number of times sequences are repeated in the training set (Carlini et al., 2020; Lee et al., 2021). We expand on this observation and quantitatively measure the effect of data duplication on memorization. Using our duplication-normalized data sample, we measure the fraction of sequences which are extractable, for buckets of sequences duplicated between 2 and 900 times. Each bucket consists of 1,000 distinct sentences, and we compute the average amount of memorization for each bucket.

Results. Figure 1b shows our results, aggregated over all sequence lengths. We observe a clear log-linear trend in memorization. While models rarely regurgitate strings that are repeated only a few times, this probability increases severely for highly duplicated strings. The small memorization values at low numbers of repetitions corroborates the positive impact of training dataset *deduplication* on memorization observed by Lee et al. (2021). However, we find that memorization does still happen, even with just a few duplicates—thus, deduplication will not perfectly prevent leakage. While this relationship is perhaps obvious, and has been corroborated for specific training examples in prior work (Carlini et al., 2019; 2020), our results show that it holds *across the entire training set*.

4.3 LONGER CONTEXT DISCOVERS MORE MEMORIZATION

The previous two questions evaluated how data collection and model training decisions impact the leakage of a model’s training data when it is provided a fixed number of tokens from a sequence as context. As a result, those experiments suggest particular actions that could be taken to mitigate memorization (by reducing model size, or limiting the number of duplicate examples).

However, even when the model is fixed, it is possible to vary the amount of extractable training data by controlling the length of the prefix passed to the model. By studying how the number of tokens of context impacts extractability, we demonstrate the difficulty of *discovering* memorization—language models may only exhibit their memorization under favorable conditions.

Results. In Figure 1c, we observe that the fraction of extractable sequences increases log-linearly with the number of tokens of context. For example, 33% of training sequences in our evaluation set are extractable from the 6B model at 50 tokens of context, compared to 65% with 450 tokens of context. We call this the **discoverability phenomenon**: some memorization only becomes apparent under certain conditions, such as when the model is prompted with a sufficiently long context.

The discoverability phenomenon may seem natural: conditioning a model on 100 tokens of context is more specific than conditioning the model on 50 tokens of context, and it is natural that the model would estimate the probability of the training data as higher in this situation. However, the result is that some strings are “hidden” in the model and require more knowledge than others to be extractable.

From one point of view, it is good that some memorization is difficult to discover. This makes it harder for attackers to perform training data extraction attacks (Carlini et al., 2020), or otherwise exploit memorization. Indeed, if an exact 100 token prompt is required to make the model output a given string, then, in practice, an adversary will likely be unable to perform the attack. The difficulty

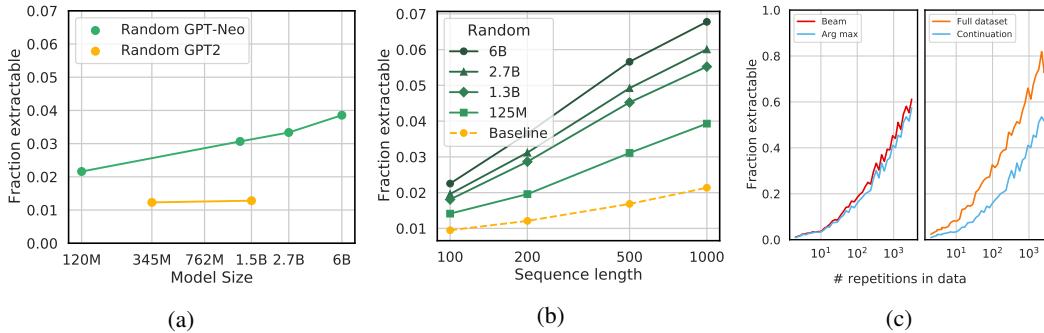


Figure 2: **(a)** Fraction of sequences extracted as a function of model scale where we sample uniformly from the training set. **(b)** Fraction of sequences extracted as we vary the length of the prompt. For each sequence length n , $n-50$ tokens are used as the prefix, and we check for extraction of the remaining 50 tokens. **(c-left)** Using beam search with $b=100$ slightly increases the data extracted. **(c-right)** We observe considerably more memorization when checking whether the generated sequence occurs anywhere in the entire training set (Section C). However, this approach is very computationally expensive so we do not use it for our other experiments.

in discovering memorization also reduces the likelihood of *non-adversarial* training data regurgitation. For example, the GitHub Copilot model (Chen et al., 2021) reportedly rarely emits memorized code in benign situations, and most memorization occurs only when the model has been prompted with long code excerpts that are very similar to the training data (Ziegler, 2021). Practitioners building language generation APIs could (until stronger attacks are developed) significantly reduce extraction risk by restricting the maximum prompt length available to users.

Viewed differently, however, the difficulty of discovering memorization can also harm our ability to audit privacy in machine learning models. Because provably-correct approaches for privacy-preserving training of machine learning models are applied only rarely in practice (Abadi et al., 2016; Thakkar et al., 2020; Ramaswamy et al., 2020), it is common to attempt post-hoc *privacy auditing* (Jayaraman and Evans, 2019; Jagielski et al., 2020; Nasr et al., 2021). Our results suggest that correctly auditing large language models likely requires prompting the model with training data, as there are no known techniques to identify the tail of memorized data without conditioning the model with a large context. Improving upon this limitation is an interesting problem for future work.

4.4 ALTERNATE EXPERIMENTAL SETTINGS

In this section, we briefly review other strategies that we could have used to quantify memorization.

Random dataset sampling. The majority of this paper uses subsets of the training data that were explicitly sampled according to training data duplication frequency. Now, we consider how our results would differ if we chose a truly random subset of the training data, where each sequence is sampled uniformly, instead of sampling a duplicate-normalized dataset. Specifically, we randomly sample 100,000 sequences of varying lengths from The Pile dataset, then prompt the model and test for memorization as before (more details in Appendix C).

Figure 2a and Figure 2b present the results. We observe similar qualitative trends with model scale and context length as in Figure 1. Larger models memorize more training examples than smaller models—and much more than the GPT-2 models that were not trained on The Pile. Similarly, providing more context to a model increases the likelihood we discover memorization. We can extract the last 50 tokens of a length-1000 sequence with 7% probability for the largest GPT-J 6B model, compared to 4% probability for the smallest 125M GPT-Neo model. (And both of these are much larger than the 2% probability of extraction for the 1.5B parameter GPT2-XL model.) These results, taken together, allow us to estimate a lower bound that there is at least 1% of The Pile dataset that is extractable by the 6B GPT-J model, but not by GPT-2 XL.

Alternate decoding strategies. We have defined memorization as a model’s ability to generate the true continuation when choosing the *most likely* token at every step of decoding. Yet, this greedy decoding strategy does not produce the overall most likely sequence. Many language model